Random Password Generator using Java OOP

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Problem Statement :

Weak and predictable passwords are a major security issue.
This project helps users generate strong and passwords easily.

Target User:

Students, general users, and anyone who wants secure password generation.

System Design & Architecture

Key Classes

- Generator → Interface for abstraction
- PasswordGenerator → Base password generator
- SecurePasswordGenerator → Strong password generator
- CharacterPool → Manages character sets
- MainUI → User interaction and GUI
- PasswordFileManager → Saves password to file

High-Level Architecture Flow

User
↓
MainUI
↓
PasswordGenerator / SecurePasswordGenerator
↓
CharacterPool
↓
Generated Password
↓
PasswordFileManager (Save)

Package Structure

- core → Password generation logic
- ui → Graphical User Interface
- util → File handling utility

OOP Concepts Used

OOP Concepts

- Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Exception Handling

Implementation

- Abstraction → Generator interface
- Encapsulation → CharacterPool
- Inheritance → SecurePasswordGenerator extends PasswordGenerator
- Polymorphism → Interface reference used
- Exception Handling → Input validation

Technical Highlight: Secure Mode

Design Decision

Secure Password Generator

A separate class SecurePasswordGenerator was created to generate stronger passwords while reusing the base logic of PasswordGenerator.

Additional Technical Features

- Input validation
- File saving
- Clipboard copy support

Why it was chosen

- Improves password strength
- Reuses existing logic
- Demonstrates inheritance
- Demonstrates polymorphism
- Keeps code modular and clean

Live Demo & Output

Demo Steps

1. Enter password length
2. Select character options
3. Enable Secure Mode
4. Click **Generate Password**
5. Click **Copy Password**
6. Click **Save Password**

Expected Output

- Random password generated
- Copy to clipboard supported
- Save to file supported

Error Handling

- Invalid input like abc
- Very short secure length like 3
- Error popup is shown

Conclusion : This project demonstrates a small but well-structured Java OOP application.

Thank You